Hafijur Raman

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Education

University of Central Florida — Orlando, FL
Master's in Computer Vision | 3.84 GPA — Expected December 2026
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Java Programming, Systems Software, Software Engineering, Advanced Computer Vision, Image Processing, Computer Graphics, Algorithms for Machine Learning, Artificial Intelligence, Data Mining.

Professional Experience

Software Engineering Intern — Waltham, MA
Xometry — June 2026 – Present
- Developing an AWS Bedrock Claude SEO agent supporting a pilot project reporting +50% quotes, +30% sessions, and +86% AI-referral CVR; planned 2 QA/monitoring agents to scale JSON-LD across 1,200+ Xometry pages and content.
- Built an object-oriented Python CLI that fetches content from Prismic and uses MongoDB/PyMongo for persistent storage, while skipping unchanged pages to prevent unnecessary Bedrock calls.

Software Engineering Fellow (Certified) — Orlando, FL
Bloomberg Tech Lab — February 2026
- Selected for Bloomberg Tech Lab and collaborated with a teammate to build a real-time messaging system using RabbitMQ.
- Developed the consumer component in Python, configured message queues, and ensured reliable message processing and acknowledgment.
- Used Git and Docker for collaboration and deployment, tested the system end-to-end, and delivered a fully functional solution on time.

Software Developer Intern — Orlando, FL
Prodisphere — June 2025 – August 2025
- Worked on an athlete evaluation algorithm using performance stats, social media data, and experience to support athlete-brand matching.
- Implemented a weighted scoring system to generate overall athlete ratings for NIL (Name, Image, Likeness) opportunities.

Software Engineer Intern — Orlando, FL
UrbanITY Lab — August 2022 – January 2023
- Developed real-time LiDAR data visualization pipelines using object-oriented C++ to improve computer vision and perception systems for autonomous vehicles.
- Designed and implemented statistical and deep learning algorithms for background filtering, object-of-interest clustering, and bounding-box regression to enhance object detection accuracy.
- Presented technical findings to research stakeholders, translating computer vision results into actionable engineering updates.

Undergraduate Teaching Assistant — Orlando, FL
University of Central Florida — January 2023 – April 2023
- Assisted Lecturer Dr. Tanvir Ahmed with a Data Structures and Algorithms course, supporting 70+ undergraduate students for 10 hours per week.
- Contributed to a 15% increase in the average passing score for students in one of UCF's most demanding computer science courses.

Projects

AI Fast Food Ordering Agent – github.com/Da-Emp66/FastFoodOrderingAgent — December 2025
- Built a system that takes voice or text commands, such as "Order me a cheeseburger," and automatically finds the nearest restaurant to start the order.
- Created a reliable scripted pipeline that extracts order details from an LLM and uses Python scripts to navigate sites like McDonald's, ensuring **100%** order accuracy.
- Architected the deployment using Docker and NGINX to host the application on a high-performance **A100** GPU server, secured with SSL for encrypted browser access.

Model Context Protocol (MCP) with Google Calendar Integration –
github.com/hafijurraman/mcp-google-calendar — June 2025

- Implemented a Node.js server using the Model Context Protocol (MCP) to provide AI applications with direct access to Google Calendar data.
- Enabled AI agents to fetch calendar events by date and return formatted meeting information through a standardized interface.

Vision Transformer (ViT)-Based ASL Recognition – github.com/hafijurraman/asl-vit January 2025 – April 2025

- Developed an American Sign Language (ASL) classifier by fine-tuning Hugging Face's pretrained Vision Transformer (ViT) model, using data augmentation and class weighting to enhance robustness.
- Achieved an overall accuracy of **94%** on a diverse dataset of over **26,000** images, demonstrating strong practical applicability for visual recognition tasks.

Pokemon GO-Inspired Web App (React, Node.js, Express, MongoDB) –
`github.com/hafijurraman/whosthatpokemon_code` April 2023 – May 2023

- Designed and developed a Pokemon GO-inspired web application using the MERN stack, integrating the Pokemon API to enable users to guess Pokemon names from images.
- Engaged more than 200 players, demonstrating the impact of the web application within the community.

Skills

- **Languages:** Python, Java, C/C++, JavaScript, SQL
- **Libraries/Frameworks:** PyTorch, TensorFlow, NumPy, Pandas, Matplotlib, React, Node.js, Express
- **Tools & Platforms:** Git, GitHub, Docker, Linux
- **Concepts:** Machine Learning, Deep Learning, Distributed Systems, Computer Vision, NLP, LLMs, Software Engineering, Backend Development, Frontend Development

Research Publications

- Salehin Seyam, Md Rashi, **Hafijur Raman**. **"Quantize the Language Tower, Not the Vision Tower: Recovering Numeric Accuracy in Page-Image QA."** Manuscript submitted to IEEE FMLDS 2026 (July 2026).
- Nusrat Jahan Prottasha, Md Kowsher, **Hafijur Raman**, Israt Jahan Anny, Prakash Bhat, Ivan Garibay, Ozlem Garibay. **"User Profile with Large Language Models: Construction, Updating, and Benchmarking."** arXiv preprint arXiv:2502.10660 (March 2025). arxiv.org/abs/2502.10660
- Nusrat Jahan Prottasha, Upama Roy Chowdhury, Shetu Mohanto, Tasfia Nuzhat, Abdullah As Sami, Md Shamol Ali, Md Shohanur Islam Sobuj, **Hafijur Raman**, Md Kowsher, Ozlem Özmen Garibay. **"PEFT A2Z: Parameter-Efficient Fine-Tuning Survey for Large Language and Vision Models."** arXiv preprint arXiv:2504.14117 (April 2025). arxiv.org/abs/2504.14117